**PROBLEM SET 6, PART 2: INTEGRALS ON MANIFOLDS**
**DUE: DEC. 14**

(1) [**Orientable manifolds**]
   (a) Prove: Any Lie group $G$ is orientable.
   (b) [not required] Suppose $M$ is orientable, $f : M \to N$ is smooth, and $q \in N$ is a regular value of $f$. Prove: $f^{-1}(q)$ is orientable.

(2) [**Orientation-preserving diffeomorphism**]
   (a) Characterize orientation-preserving diffeomorphism via volume forms.
   (b) Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be the map $f(x) = -x$. Is $f$ orientation-preserving?
   (c) Let $f : S^n \to S^n$ be the antipodal map $f(x) = -x$. Is $f$ orientation-preserving?
   (d) Prove: For any $k$, $\mathbb{RP}^{2k}$ is not orientable.

(3) [**An integral**]
Consider the 2-form

$$\omega = xdy \wedge dz + y^2 dz \wedge dx + zdx \wedge dy$$

on $\mathbb{R}^3$. Let $\iota : S^2 \to \mathbb{R}^3$ be the standard embedding. Compute the integral $\int_{S^2} \iota^* \omega$.

(4) [**The divergence of a smooth vector field**]
Let $M$ be a smooth manifold, $\mu$ a volume form on $M$, and $X$ a smooth vector field on $M$. Then the Lie derivative $\mathcal{L}_X \mu$ is again a smooth $n$-form on $M$. The *divergence* of $X$ with respect to the volume form $\mu$ is the function $\mathrm{div}_\mu(X)$ such that

$$\mathcal{L}_X \mu = \mathrm{div}_\mu(X)\mu.$$

   (a) Find the expression of $\mathrm{div}_\mu(X)$ in local coordinates.
   (b) Prove: For any $f \in C^\infty(M)$,

$$\mathrm{div}_\mu(gX) = g\mathrm{div}_\mu(X) + X(g)\mu.$$

   (c) Prove **Gauss formula**:

$$\int_M (\mathrm{div}_\mu X)\mu = \int_{\partial M} \iota_X \mu.$$

   (d) We say a vector field $X$ is *divergence free* if $\mathrm{div}_\mu(X) = 0$. Prove: If $M$ is a compact manifold without boundary (i.e. $\partial M = \emptyset$), and $X$ is a divergence free vector field on $M$, then for any $f, g \in C^\infty(M)$, one has

$$\int_M X(f)g\mu = -\int_M fX(g)\mu.$$

   (In other words, the linear operator $X : C^\infty(M) \to C^\infty(M)$ is skew-symmetric.)

(5) [**The transport equation**] [not required]
Let $M$ be a smooth manifold, $\mu$ a volume form on $M$, and $\Omega \subset M$ an open set in $M$ with smooth boundary. Let $X$ be a complete vector field on $M$, and $\Omega_t = \phi_t(\Omega)$ the flow-out of $\Omega$ by the flow $\phi_t$ generated by $X$.

(a) For any $\rho \in C^\infty(\mathbb{R} \times M)$ (as usual we denote $\rho_t = \rho(t, \cdot) \in C^\infty(M)$), prove:

$$\frac{d}{dt} \int_{\Omega_t} \rho_t \mu = \int_{\Omega_t} \left( \frac{\partial \rho}{\partial t} + \mathrm{div}_\mu(\rho_t X) \right) \mu. \qquad \text{[The transport equation]}$$

(b) Let $\rho$ be the mass density of an ideal fluid moving in a compact region. Assume the *conservation of mass*, i.e. for any region $\Omega$, $\int_{\Omega_t} \rho_t \mu$ is independent of $t$. Prove:

$$\frac{\partial \rho}{\partial t} + \mathrm{div}_\mu(\rho_t X) = 0. \qquad \text{[The equation of continuity]}$$